Reproducible Astrophysics with Nix

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For those who want to follow the optional hands-on activity, please install
Nix: the package manager
from nixos.org/download
onto your system.
If you want a headstart, all the activities are located in:
`https://github.com/Vasissualiyp/Nix-Tutorial`

You will need a terminal for this activity!

Astrophysics' Reproducibility Challenge

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¯_(ツ)_/¯
IT WORKS
on my machine

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All fail at bit-for-bit reproducibility across systems/time

Nix: A Paradigm Shift

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A system that focuses on construction of **declarative and reproducable** software environments.

Why Nix?

- **Pure (Isolated) Builds**

  No host leaks - strict sandboxing

- **Cryptographic Hashing**

  Unique IDs for *complete* dependency trees

- **Declarative Environments**

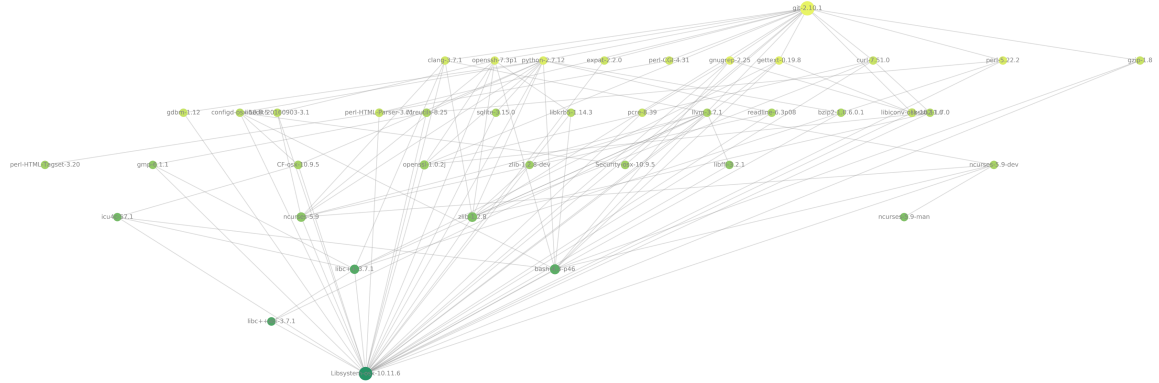
  Define what you want, not how to build it

- **Atomic Operations**

  Instant rollbacks, no broken states

Why build environment isolation?

Dependency graph:



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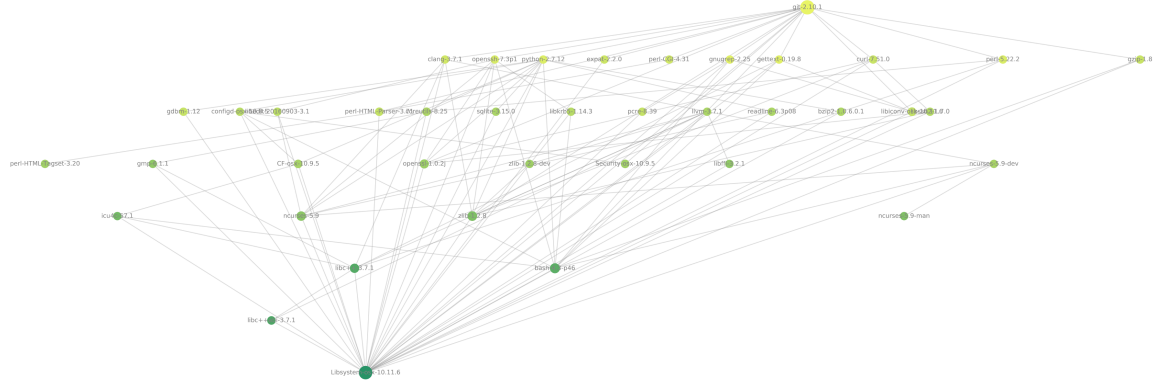

Environment purity (isolation) prevents dependency on an external (host, internet) state!

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This means that if 2 builds of some package have the same hash, they are guaranteed to be built in an identical way! $\rightarrow$ **functionally identical builds**

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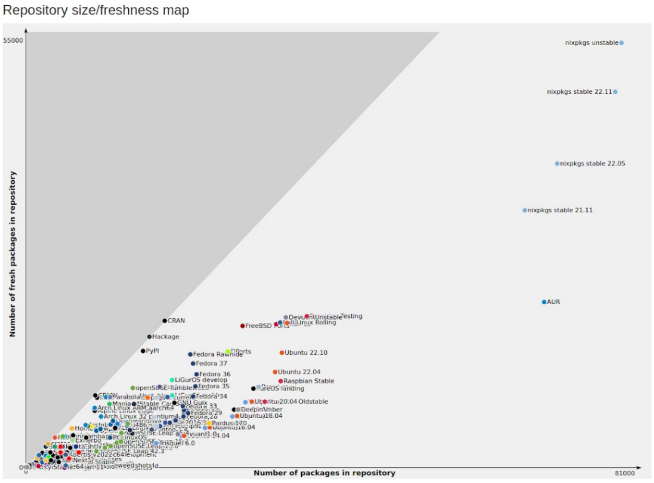

Nixpkgs

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When you are creating your environment, you usually pull the packages and dependencies from nixpkgs.

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- Pre-Nix: could only run on Niagara
- Post-Nix: reproducible on any machine with Nix

Dependencies include:

- GCC (GNU Compilers for C, Fortran)
- GSL (GNU Scientific Library)
- OpenMP (Parallelism)
- OpenMPI (Parallelism)
- FFTW (both single and double precision with MPI support)
- HDF5
- And 10+ Python packages (Including nontrivially-packaged)

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All can be activated on any Nix-enabled machine with a single command!

```
{
  description = "PeakPatch developement environment";

  inputs = {
    nixpkgs.url = "github:NixOS/nixpkgs/nixos-unstable";
    flake-utils.url = "github:numtide/flake-utils";
  };
  outputs = { self, nixpkgs, flake-utils, ... }:
    flake-utils.lib.eachDefaultSystem (system:
      let
        pkgs = import nixpkgs {
          inherit system;
          config.allowUnfree = true;
        };
        customFftw_single = # ... package override (for MPI)
        customFftw_double = # ... package override (for MPI)
        python = pkgs.python311Packages;
        camb = python.buildPythonPackage rec {
          # package declarations (not in nixpkgs)
        };
        class = python.buildPythonPackage rec {
          # package declarations (not in nixpkgs)
        };
    # ... More packages ...
```

```
        pythonEnv = (python.python.withPackages (ps: with ps; [
          pandas
          matplotlib
          numpy
          astropy
          # ... More python packages
        ]));
      in
      {
        devShell = pkgs.mkShell {
          buildInputs = with pkgs; [
            pythonEnv
            gsl
            gcc
            llvmPackages.openmp
            customFftw_single
            customFftw_double
            # ... More packages
          ];
          shellHook = ''
            export PP_DIR=$(pwd)
      # ... More environment variables and welcome message
            '';
        };
      }
    );
}
```

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Example:

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Example:

Your laptop physically broke, but all your data is backed up.
NixOS: Re-install your system on a new machine with your config file

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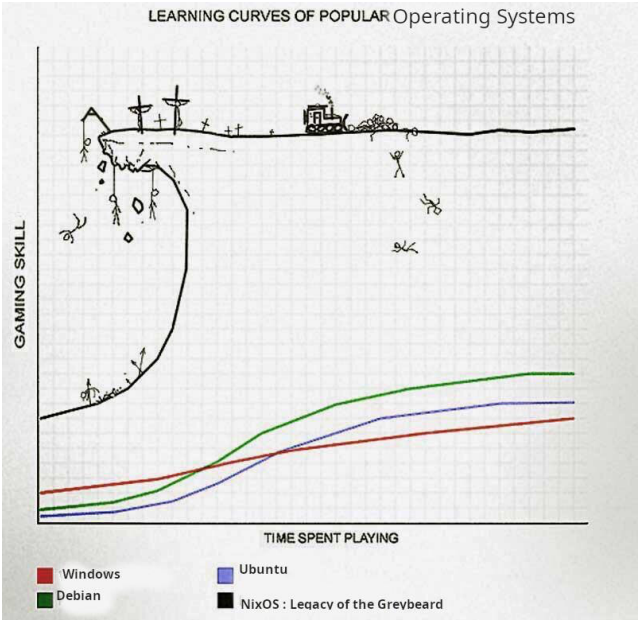
If you are ready to pay the price...


LEARNING CURVES OF POPULAR Operating Systems
GAMING SKILL
TIME SPENT PLAYING
Windows
Debian
Ubuntu
NixOS : Legacy of the Greybeard

When to Use Nix (and When Not To)

Strengths:

- Long-term reproducibility
- HPC environment control
- Multi-project collaboration
- Cross-platform consistency
- No dependency hell

Challenges:

- Learning curve
- Proprietary software ($\pm$)
- IDE integration ($\pm$)
- No POSIX compliance by default (It's actually good)
- Storage overhead (10-50GB) (depending on setup)

Getting Started

- zero-to-nix.com
- nixos.org
- nixos.org/guides/nix-pills
- discourse.nixos.org

Conclusion

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- Essential for long-term astrophysics projects
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Thank You!

References I

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